Entailment-Driven Privacy Policy Classification with LLMs

Bhanuka Silva

BuildSEC'24 Building a Secure & Empowered Cyberspace
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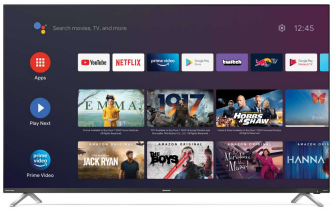
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Hyper-connected World

Today, any device that is connected to internet would likely require some account login where our personal information will be collected or transmitted.







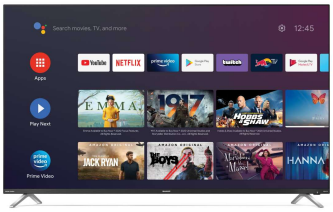


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Hyper-connected World

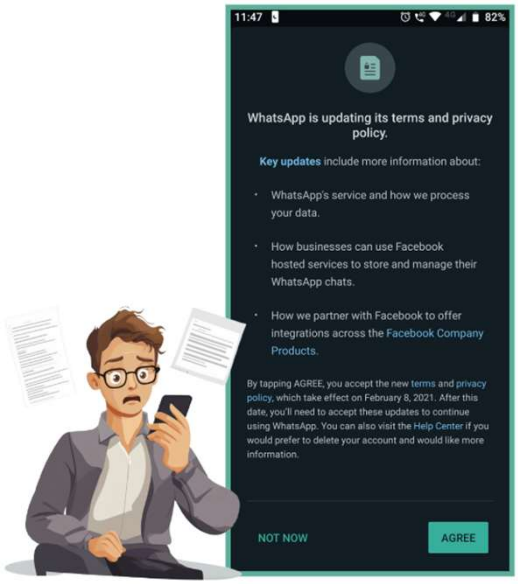
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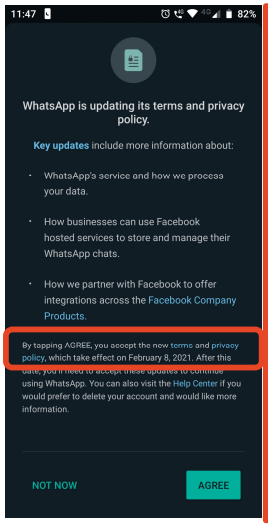


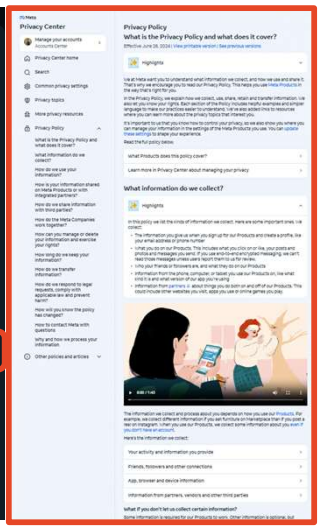

Collection, use or disclose of personal information is often justified by individual's consent

Privacy Paradox?



Uninformed Consenting





- Too Long
- Too Complex
- Legal Jargon

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Literature: Automated Methods to Provide Meaningful Interpretations

- Identifying privacy topics / labels [Liu et al 2018], [Srinath et al. 2020, 2021],
- Summarising sections [Gopinath et al. 2020],
- Highlighting informed user-choices (e.g., opt-out choices [Sathyendra et al. 2016, 2017]),
- Question-answering systems [Harkous et al. 2018],
- User-interface tools to identify common data practices (e.g., browser extensions [Zimmect et al. 2014]),
- Detecting privacy policy inconsistencies and non-compliances [Linden et al. 2018], [Story et al. 2019].

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They either rely on traditional natural language processing techniques,
Or do not provide sufficient explainability behind the interpretations.

LLMs for Policy Classification?

(a) Personally Identifiable Information. Personally Identifiable Information refers to information by which you may be personally identified, such as your name, email address, or school. We will collect and process such data about you when you register for a Service/create an account, subscribe to a newsletter, submit feedback, enter a contest, fill out a survey, or send us a communication. When ordering products or services on the Websites or App, you may also be asked to provide a credit card number and other payment-related information. You may also provide information to be published or displayed (posted) on public areas of the Websites (such as blogs, forums, message boards, and the like), or transmitted to other users of the Websites using sharing features (collectively, User Contributions). Your User Contributions are posted and transmitted to others at your own risk. We cannot control the actions of other users of the Websites with whom you may choose to share your User Contributions. Depending upon the activity, some of the information we ask you to provide is identified as mandatory and some as voluntary. If you do not provide the mandatory data with respect to a particular activity, you will not be able to engage in that activity.

Does this privacy policy paragraph contain information about **data retention**?

The description **doesn't directly state** how long the information is stored, but the context of requiring certain information for activity engagement suggests that data retention policies are in place, even though they're not **explicitly** outlined.

A Matter of Alignment and Logical Reasoning

- **Privacy Policies**
  - Legal text
  - High information density and longer context
- **GPT Outputs**
  - Human-centric conversation style
  - Hallucinations
  - Lack of Reasoning

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A Matter of Alignment and Logical Reasoning

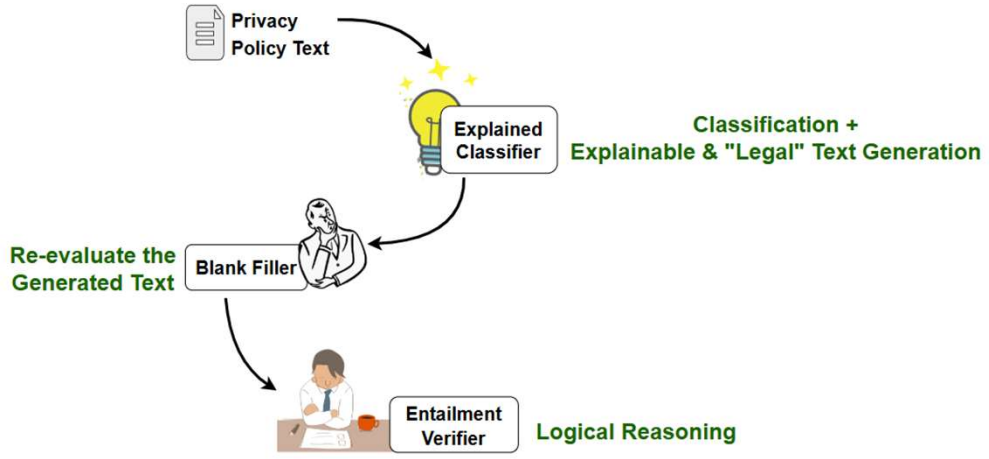
- **Privacy Policies**
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"Are you willing to copy paste a GPT output without reading it first ?"

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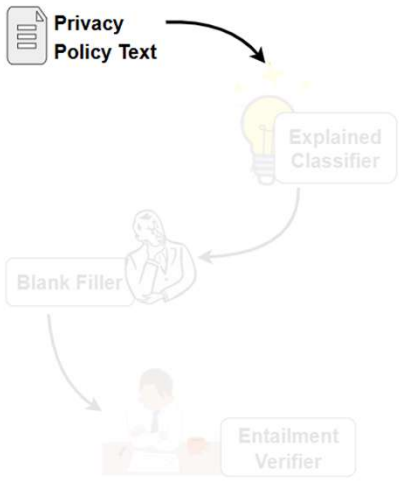
Proposed Framework: Entailment-driven Privacy Policy Classification



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Our Framework

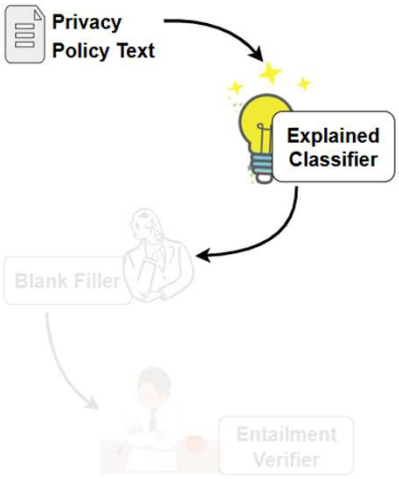


"…..We retain indefinitely all the information we gather about you in an effort to make your repeat use of our website more efficient, practical, and relevant...."

12

12

Our Framework



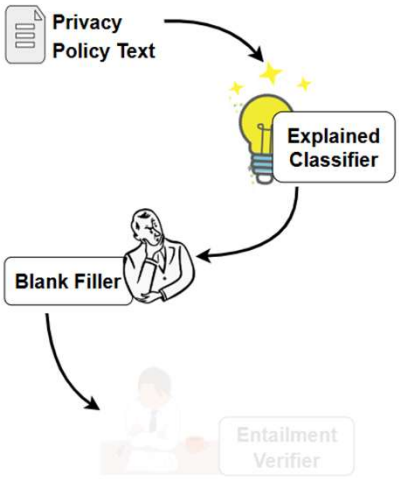
"....We retain indefinitely all the information we gather about you in an effort to make your repeat use of our website more efficient, practical, and relevant...."

Classification: **Data Retention**
Reasoning: Paragraph mentions "**we retain indefinitely**"

13

13

Our Framework



"....<MASK> all the information we gather about you in an effort to make your repeat use of our website more efficient, practical, and relevant...."

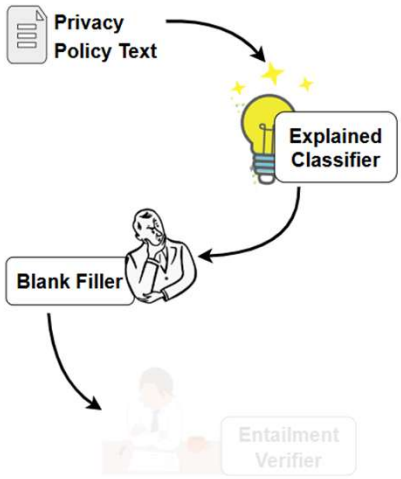
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MASK : ?

14

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Our Framework



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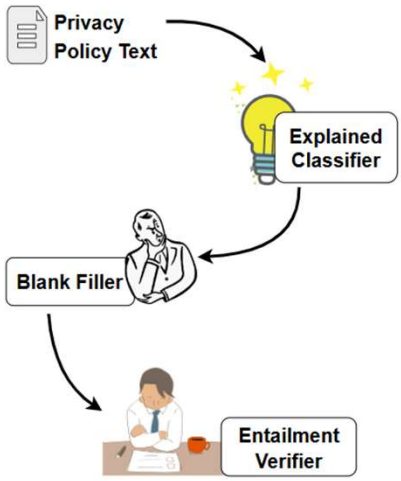
Classification: **Data Retention**

MASK : "we will maintain"

15

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Our Framework



"....We retain indefinitely all the information we gather about you in an effort to make your repeat use of our website more efficient, practical, and relevant...."

Classification: **Data Retention**
Reasoning: Paragraph mentions "**we retain indefinitely**"

MASK : "we will maintain"

Prediction: "Entailment"

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Dataset

The OPP-115 corpus (Online Privacy Policies, set of 115) is a collection of website privacy policies. (Wilson, S. et al., 2016)

"If you contact us by phone or email, we may keep a record so that we can follow up with you if necessary. We do not disclose the information in that record to any outside entity unless you give us permission to do so or unless disclosure is compelled by law."

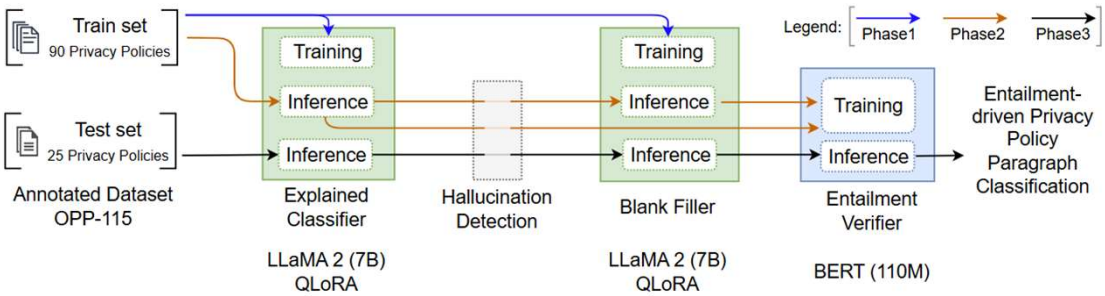
Data Retention*
Third Party Sharing/Collection*



Each paragraph could contain multiple data-practices, therefore, a multi-label multi-class classification.

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Model Training



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Results: Compared with Generative Benchmarks

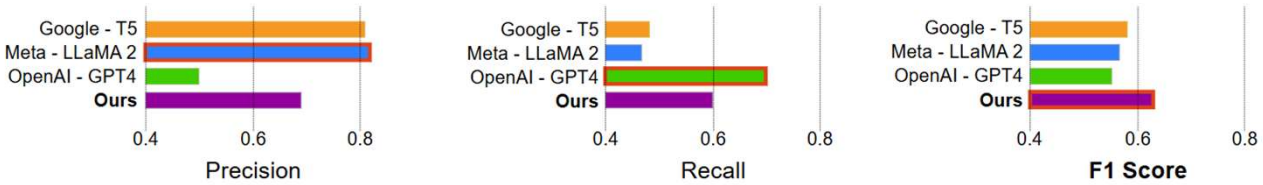
Class	T5 P	T5 R	T5 F1	GPT4 Prompting P	GPT4 Prompting R	GPT4 Prompting F1	LLaMA2 7B Fine-tuned P	LLaMA2 7B Fine-tuned R	LLaMA2 7B Fine-tuned F1	Our Method P	Our Method R	Our Method F1	Support
First Party Collection/Use	0.86	0.78	0.82	0.68	0.91	0.78	0.92	0.66	0.77	0.83	0.72	0.77	289
Third Party Sharing/Collection	0.87	0.65	0.74	0.70	0.76	0.73	0.92	0.48	0.63	0.82	0.78	0.80	204
User Choice/Control	0.75	0.43	0.55	0.47	0.71	0.57	0.87	0.39	0.54	0.62	0.67	0.64	115
User Access, Edit and Deletion	0.86	0.61	0.71	0.47	0.80	0.59	0.83	0.49	0.62	0.67	0.70	0.68	41
Introductory/Generic	0.72	0.43	0.54	0.69	0.31	0.43	0.43	0.75	0.55	0.56	0.44	0.50	118
Policy Change	0.93	0.48	0.64	0.43	0.79	0.55	1.00	0.48	0.65	0.54	0.56	0.55	29
Data Security	0.89	0.52	0.65	0.52	0.77	0.62	0.94	0.48	0.64	0.83	0.69	0.75	62
International & Specific Audience	0.86	0.70	0.77	0.64	0.87	0.74	0.73	0.72	0.72	0.64	0.67	0.66	60
Practice Not Covered	0.27	0.03	0.05	0.44	0.25	0.32	0.33	0.01	0.02	0.48	0.33	0.39	110
Data Retention	0.75	0.12	0.20	0.38	0.46	0.41	1.00	0.19	0.32	0.59	0.50	0.54	26
Privacy Contact Information	1.00	0.39	0.56	0.49	0.75	0.59	1.00	0.41	0.58	0.73	0.75	0.74	56
Do Not Track	1.00	0.60	0.75	0.15	1.00	0.26	1.00	0.60	0.75	1.00	0.40	0.57	5
Micro Average	0.83	0.54	0.66	0.58	0.70	0.63	0.77	0.50	0.61	0.72	0.64	**0.68**	1115
Macro Average	0.81	0.48	0.58	0.50	0.70	0.55	0.83	0.47	0.57	0.69	0.60	**0.63**	1115
Weighted Average	0.79	0.54	0.62	0.60	0.70	0.62	0.80	0.50	0.58	0.71	0.64	**0.67**	1115

: F1 score < 50% : Highest in a row

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Results: Compared with Generative Benchmarks



Google T5: -8.6%
Meta LLaMA2: -14.5%
OpenAI–GPT4: -10.5%

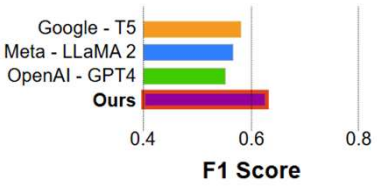
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Ablation Study

ABLATION STUDY: ALL VALUES ARE MACRO-AVERAGE SCORES

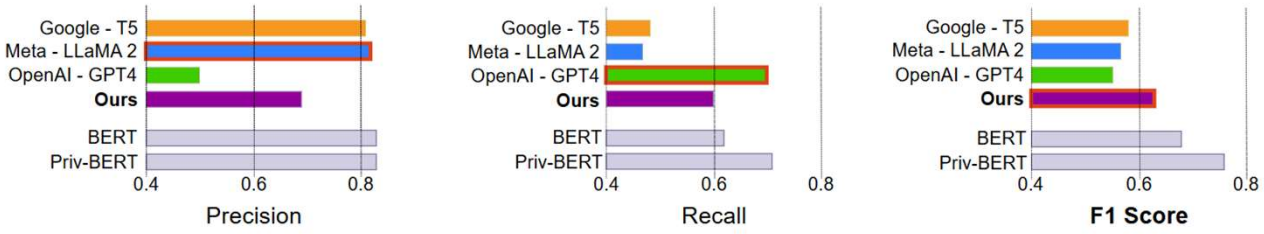
Description	P	R	F1
Explained classifier only	0.38	0.85	0.48
Explained classifier + entailment verifier	0.61	0.61	0.59
Full pipeline	0.69	0.60	0.63



We highlight that even without the blank filler, our model's macro-averaged F1 score is higher than zero-shot GPT4 and finetuned LLaMA2.

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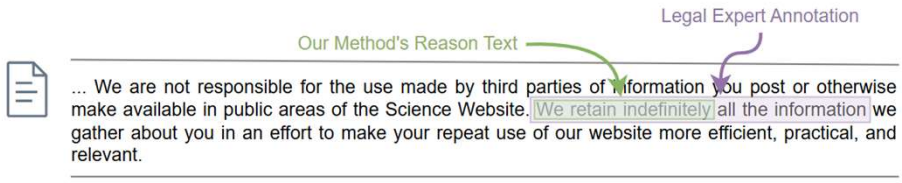
Results: Compared with Generative and Encoding Benchmarks



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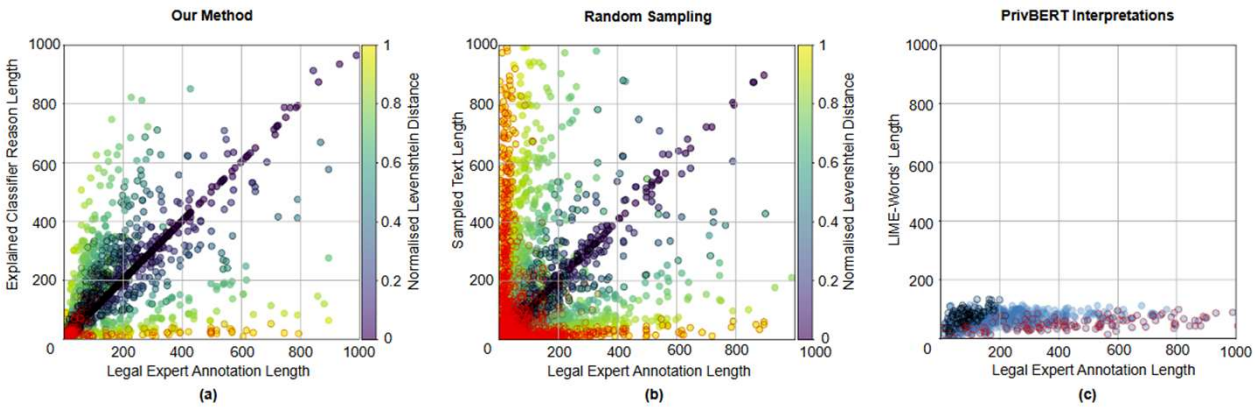
Explainability

Explained classifier answers the explanatory linguistic question of 'why?' a particular class label is relevant:

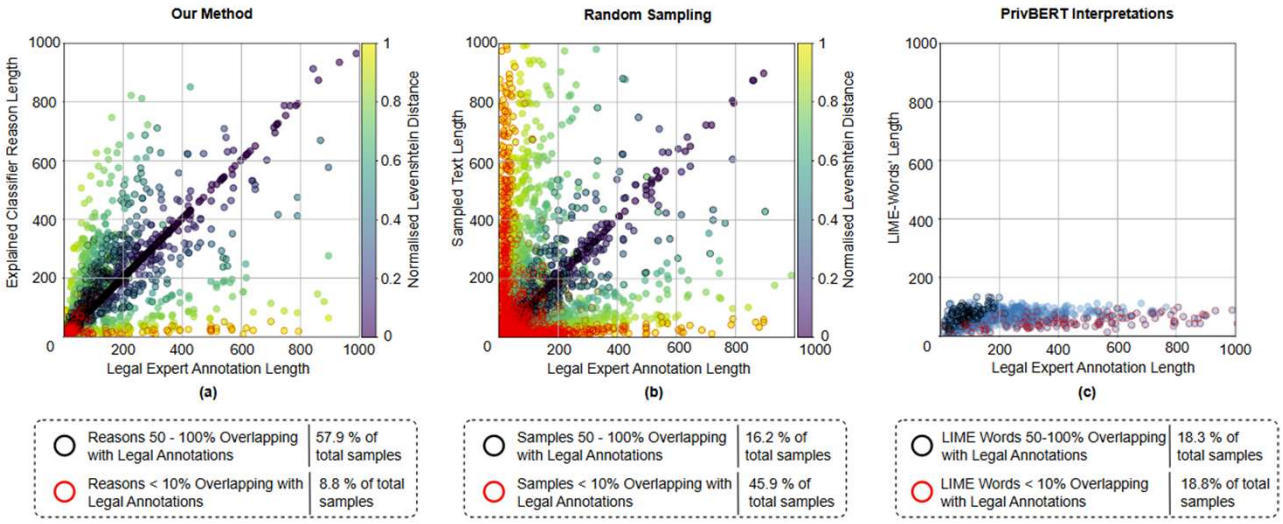


a) Normalised Levenshtein Distance
b) Overlap Percentage (Jaccard similarity)

Alignment with Legal Expertise



Alignment with Legal Expertise



Can Legal Experts Make Mistakes Too?

"B. Targeted Advertising Companies Everyday Health works with may use cookies for purposes including understanding Web usage patterns of users who see advertisements on the Sites, controlling the sequence of advertisements you see, making sure you don't see the same advertisement too many times, determining your interests and tailoring advertisements and promotions based on those interests."

- **What would be the best classification label?**

Can Legal Experts Make Mistakes Too?

"B. Targeted Advertising Companies Everyday Health works with may use cookies for purposes including understanding Web usage patterns of users who see advertisements on the Sites, controlling the sequence of advertisements you see, making sure you don't see the same advertisement too many times, determining your interests and tailoring advertisements and promotions based on those interests."

- **Legal Expert:** First Party Collection/Use
- **Our Method:** Third Party Sharing /Collection

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Summary

- We propose an entailment-driven LLM-based framework for privacy policy paragraph classification.
- Our method performs better than vanilla LLM-methods; ~11% higher.
- Our method generates explanations behind those predictions, aligned with what a legal expert would have reasoned.

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Summary

- We propose an entailment-driven LLM-based framework for privacy policy paragraph classification.
- Our method performs better than vanilla LLM-methods; ~11% higher.
- Our method generates explanations behind those predictions, aligned with what a legal expert would have reasoned.

While LLMs can provide user-friendlier means to access privacy policies, they are not that useful in their vanilla form.
Rather, it is necessary to have auxiliary steps as we propose in our framework.

Thank You

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